A squarefree floor sum and its class-number inversion

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Abstract

We prove the odd-squarefree floor-sum identity of Chamberland and Dilcher and its equivalent Möbius-inversion formula. Expanding the number of square roots modulo a squarefree integer into quadratic characters gives an exact remainder sum; an elementary lattice count then recovers the floor sum. The two equivalent source conjectures receive one campaign credit.

Task 1: the original conjecture

For a positive integer n, define

$$f(n) = \sum_{j=1}^{\lfloor n/4 \rfloor} \left\lfloor \sqrt{jn} \right\rfloor - \frac{n^2-1}{12}.$$

For odd squarefree d, let $\delta(d) = 1$ when $d \equiv 3 \pmod 4$ and $\delta(d) = 0$ otherwise. Define $H(d) = 0$ in the latter case, and

$$H(3) = \frac{1}{3}, \qquad H(d) = h\big(\mathbb{Q}(\sqrt{-d})\big) \quad (d > 3,\ d \equiv 3 \pmod 4).$$

Thus $H(d)$ is the source's product $h^*(-d)\delta(d)$, with no need to assign an unused normalization when $\delta(d) = 0$.

Theorem 1. *For every odd squarefree positive integer n,*

$$f(n) = \frac{1-n}{4}\delta(n) - \frac{1}{2}\sum_{d \mid n} H(d), \tag{1}$$

$$H(n) = \sum_{d \mid n} \mu(n/d)\left(\frac{1-d}{2}\delta(d) - 2f(d)\right). \tag{2}$$

Equation (2) is Conjecture 7.4 of [1, p. 22], MathDB 370788. Equation (1) is Conjecture 7.3, MathDB 370787. Möbius inversion makes them equivalent; we count them as one resolution. The case $n = 1$ has only empty sums and is immediate.

Counting square roots and evaluating the remainder sum

For odd squarefree $d > 1$, write $\chi_d(t) = (t/d)$ for the Jacobi symbol; put $\chi_1(t) = 1$. This is a nonprincipal quadratic character modulo d when $d > 1$. Its character sum is zero. One elementary

way to see this is to choose, by the Chinese remainder theorem, a unit that is a quadratic nonresidue at exactly one prime dividing d; multiplication by this unit negates the sum.

The number $\rho_n(t)$ of solutions of $x^2 \equiv t \pmod{n}$ is

$$\rho_n(t) = \prod_{p|n} \left(1 + \left(\frac{t}{p}\right)\right) = \sum_{d|n} \chi_d(t). \tag{3}$$

This includes nonunit t: modulo a prime there is one root at zero and zero or two roots at a nonzero residue. The Chinese remainder theorem then proves the product for every t.

Define

$$S_n = \frac{1}{n} \sum_{x=0}^{n-1} (x^2 \bmod n).$$

Interchanging finite sums in (3) gives

$$S_n = \sum_{d|n} \frac{1}{n} \sum_{t=0}^{n-1} t\chi_d(t).$$

The contribution of $d = 1$ is $(n-1)/2$. For $d > 1$, split t into complete periods modulo d. Since the unweighted character sum is zero,

$$\frac{1}{n} \sum_{t=0}^{n-1} t\chi_d(t) = \frac{1}{d} \sum_{r=0}^{d-1} r\chi_d(r).$$

If $d \equiv 1 \pmod{4}$, the character is even, and pairing r with $d - r$ shows that this weighted sum is zero. If $d \equiv 3 \pmod{4}$, quadratic reciprocity identifies $\chi_d(r)$ with the Kronecker character $(-d/r)$. The discriminant $-d$ is fundamental, so Dirichlet's finite class-number formula gives

$$\frac{1}{d} \sum_{r=0}^{d-1} r\chi_d(r) = -H(d).$$

Here we use the classical formula stated in [1, Theorem 4.1]; its unit factor is six at $d = 3$ and two otherwise, giving exactly $H(3) = 1/3$. Consequently

$$S_n = \frac{n-1}{2} - \sum_{d|n} H(d). \tag{4}$$

Recovering the floor sum

We include the lattice-count step to fix every endpoint. Put $\nu = \lfloor n/4 \rfloor$ and $F(n) = f(n) + (n^2 - 1)/12$. For odd $n \geq 3$ one has $\lfloor \sqrt{\nu n} \rfloor = 2\nu$, including $\nu = 0$. Counting the pairs (j, k) with $1 \leq j \leq \nu$ and $k^2 \leq jn$ gives

$$F(n) = \sum_{k=1}^{2\nu} \left(\nu + 1 - \left\lceil \frac{k^2}{n} \right\rceil\right) = 2\nu^2 - \sum_{k=1}^{2\nu} \left\lfloor \frac{k^2}{n} \right\rfloor.$$

The last equality uses squarefreeness: $n \nmid k^2$ for $0 < k < n$. Writing $R_n(k) = k^2 \bmod n$ and summing the consecutive squares yields

$$F(n) = 2\nu^2 - \frac{2\nu(2\nu+1)(4\nu+1)}{6n} + \frac{1}{n} \sum_{k=1}^{2\nu} R_n(k). \tag{5}$$

Since $R_n(n-k) = R_n(k)$,

$$\frac{1}{n}\sum_{k=1}^{(n-1)/2} R_n(k) = \frac{S_n}{2}.$$

If $n \equiv 1 \pmod 4$, then $2\nu = (n-1)/2$. If $n \equiv 3 \pmod 4$, then $2\nu = (n-3)/2$, and the omitted last remainder is

$$R_n\left(\frac{n-1}{2}\right) = \frac{n+1}{4},$$

as follows by subtracting $(n+1)/4$ from $(n-1)^2/4$. Substituting these two cases into (5) and subtracting $(n^2-1)/12$ gives respectively

$$f(n) = \frac{S_n}{2} - \frac{n-1}{4}, \qquad f(n) = \frac{S_n}{2} - \frac{n-1}{2}.$$

Equation (4) now proves (1).

Finally, multiply (1) by -2 and rearrange:

$$\frac{1-n}{2}\delta(n) - 2f(n) = \sum_{d|n} H(d).$$

Every divisor of an odd squarefree integer is again odd and squarefree. Ordinary Möbius inversion on the divisor set therefore gives Equation (2), completing the proof.

Tasks 2–5: follow-up questions

2. Extend the root-count expansion to arbitrary odd nonsquarefree integers and obtain explicit coefficients for all class-number terms.

3. Derive effective uniform bounds for the deviation of $f(n)$ from its residue-class main term over odd squarefree integers.

4. Replace the linear remainder weight by higher powers and express the resulting sums using generalized Bernoulli numbers or Dirichlet values.

5. Compare algorithms for computing the class numbers in Equation (2), including their exact arithmetic complexity, against enumeration of reduced binary quadratic forms.

Verification and scope. The verifier computes the original floor and remainder sums using integer arithmetic and computes class numbers independently by enumerating reduced binary quadratic forms. It checks both identities and the inversion. Finite checks corroborate the all-squarefree proof. This is author-agent review, not independent peer review. Source and search limitations are recorded in the literature audit.

References

[1] M. Chamberland and K. Dilcher, *Sums of the floor function related to class numbers of imaginary quadratic fields*, arXiv:2510.04387v1 (2025), Conjectures 7.3–7.4, pp. 21–22. All 23 pages of the current manuscript were read on 29 September 2026.

[2] MathDB, problem 370788 and the equivalent problem 370787, live records checked 29 September 2026.